Isolated Tears of the Anterior Cruciate Ligament

Over 30-Year Follow-up of Patients Treated With Arthrotomy and Primary Repair

COL (ret) Dean C. Taylor,*† MD, CPT Matthew Posner,‡ MD, COL (ret) Walton W. Curl,§ MD, and COL (ret) John A. Feagin,‖ MD
From the †Duke Sports Medicine Center, Duke University Medical Center, Durham, North Carolina, ‡William Beaumont Army Medical Center, El Paso, Texas, §Department of Orthopaedic Surgery and Sports Medicine, Wake Forest University Medical Center, Winston-Salem, North Carolina, and ‖Vail, Colorado

Background: Over 30 years ago, Feagin and Curl reported on the diagnosis and treatment of "isolated" injuries of the anterior cruciate ligament (ACL). The purpose of this study was to evaluate the long-term results of the patients initially diagnosed with tears of the ACL, with special emphasis on those treated with primary repair.

Hypothesis: Long-term results are unsatisfactory for open evaluation and treatment of ACL injuries with or without primary repair.

Study Design: Case series; Level of evidence, 4.

Methods: Follow-up was available on 34 of the original 57 patients (60%) who had ACL tears on evaluation by arthrotomy. The average age at the index procedure was 20 years, and average follow-up was 32 (range, 29-36) years. Twenty-five of the 26 complete ACL tears were treated with primary repair. None of the 8 partial tears was repaired. Evaluations included the subjective and symptom ratings of the International Knee Documentation Committee (IKDC) evaluation form, Lysholm score, Knee Injury and Osteoarthritis Outcome Score (KOOS), Tegner activity scale, and SANE rating.

Results: For the IKDC subjective score, 18 of 34 patients rated their knees as normal or nearly normal. The Lysholm scores averaged 70.1 (range, 20-100). The average SANE rating at 5 years was 74.8 (range, 49-110), and the current SANE rating averaged 68.9 (range, 10-100). The overall KOOS rating averaged 68.6 (range, 26.4-100). The Tegner activity score was 3.7. There was little difference between patients with unrepaired partial tears and those who had primary repairs.

Conclusion: At more than 30-year follow-up, patients have decreased activity levels and an equal mix of acceptable and unacceptable outcomes. We were unable to identify any predictive factors that correlated with the results; however, subsequent meniscal surgery did correlate with poor results. The results at greater than 30 years reinforce the 5-year results that showed unsatisfactory results after the open evaluation and treatment of ACL injuries with or without repair.

Keywords: ACL reconstruction; primary repair; knee

*Address correspondence to COL (ret) Dean C. Taylor, MD, Duke University Medical Center, Duke Sports Medicine Center, Box 3615, Durham, NC 27710 (e-mail: dean.taylor@duke.edu).

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In 1972, Feagin, Abbott, and Rokous presented their findings in knee injuries treated at the United States Military Academy at West Point.[5] They described a combination of events that we now recognize as an injury history consistent with an anterior cruciate ligament (ACL) tear. They performed 64 arthrotomies in patients who had 4 essential elements: a history of a deceleration-twisting injury to the knee, a sensation of a pop within the knee joint at the time of injury, an inability to continue playing after the injury, and the development of a tense knee effusion in the 24

hours after injury. Fifty-seven of the 64 knees had injuries to the ACL. They reported that at 2-year follow-up, 25 of 30 patients had good or excellent results.

In 1976, Feagin and Curl published a 5-year follow-up study of 32 of the original patients who had primary repair of isolated tears of the anterior cruciate ligament.[6] Contrary to the 1972 report, the vast majority of patients were symptomatic with impaired athletic activities. They concluded that "the anterior cruciate ligament is critical to the function of the knee in the young athlete" and that "an injury to the anterior cruciate ligament is a compromising event even when subjected to repair."

In the 30 years since the first West Point reports, ACL tears have become a well-recognized orthopaedic injury. Surgical treatment has evolved from the initial attempts at primary repair to reconstructions that are now one of the most commonly performed orthopaedic operations.[8] Changes in surgical treatment have improved the short-term treatment results and presumably will improve long-term results. The purpose of this study was to provide the long-term results on the patients studied by Feagin and Curl. Although not a commonly performed procedure currently, these results can be compared subsequently to other procedures once adequate time has passed to provide long-term outcomes on the other procedures. We hypothesized that long-term results after open evaluation and treatment of the ACL with or without primary repair would continue to be unsatisfactory.

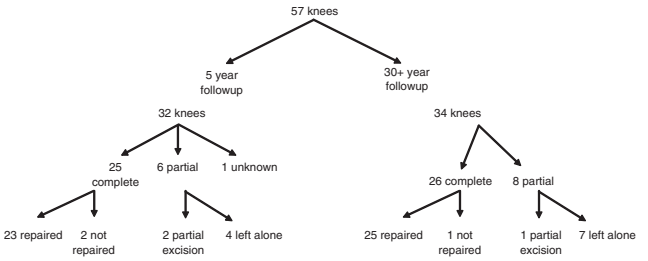


Figure 1. Flowchart of the patient population.

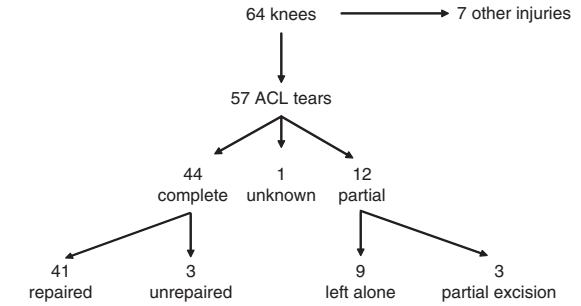


Figure 2. Flowchart of the 64 knees initially reported by Feagin et al.[5]

MATERIALS AND METHODS

Sixty-two patients (64 knees) who underwent arthrotomies for acute hemarthrosis with the presumptive diagnosis of ACL injuries between 1964 and 1970 were identified from the records that were used for the 1972 and 1976 reports. Figure 1 is a flowchart for evaluations on the original 64 knees. Fifty-seven patients had ACL injuries diagnosed at arthrotomy; these 57 were the study population for this investigation. Figure 2 outlines the follow-up for the original patients reported by Feagin et al.[5]

All patients were male United States Military Academy cadets or graduates (at the time, only males were enrolled at the United States Military Academy). The original operative notes and narrative summaries from the hospitalizations were available for review. Questionnaires used for the 5-year study were available for an incomplete number of patients. One question used at the 5-year follow-up questionnaire asked of each patient was the following: "If your knee was 100% before injury, what would you rate it now?" We used the responses to this question as an equivalent to the Single Assessment Numeric Evaluation (SANE) rating.[20]

Two of the original 57 patients are deceased, and 34 patients responded to our current query to participate in this project. Patients were interviewed by mail, telephone, or in person (while attending a West Point reunion). Twenty-three of these 34 participated in Feagin and Curl's 5-year follow-up study.[6] The average age at the index procedure was 20 (range, 18-36 years). One patient was an officer (36 years old); the remaining patients were cadets (aged 18-22). The average age at follow-up was 52 (range, 47-67 years), for an average follow-up of 32 years (range, 29-36 years).

The most common sport of injury was football (n = 15), followed by lacrosse (n = 9). There were 2 injuries in military training, 2 in basketball, and 1 each in skiing, wrestling, soccer, gymnastics, judo, and volleyball. Twenty of the 34 injuries (59%) were noncontact injuries. All lacrosse injuries were noncontact, and 7 of the 15 football injuries were noncontact injuries.

The index procedure was substantially different from current methods. The average time from injury to surgery was 10 days (range, 2-30 days). The operations were performed under general anesthesia in 31 cases and spinal anesthesia in 3 cases. The average operative time was 64 minutes (range, 30-138 minutes), and patients were hospitalized for an average of 13 days (range, 7-23 days) with many patients being hospitalized for several days before surgery. Fourteen of the 34 patients had Buck traction before surgery while in the hospital. The average time from the date of surgery to discharge from the hospital was 9 days (range, 5-20 days).

All patients had arthrotomies of the knee. The original records show that 26 of the 34 had complete ACL tears and the remaining 8 had partial tears. Twenty-four of the complete tears were from the femoral attachment of the ACL, and 2 were midsubstance. Two patients also had chondral injuries that were treated by debridement of the lesion—one in a patient with a partial interstitial tear, and one in a patient with a complete proximal ACL tear. No patient

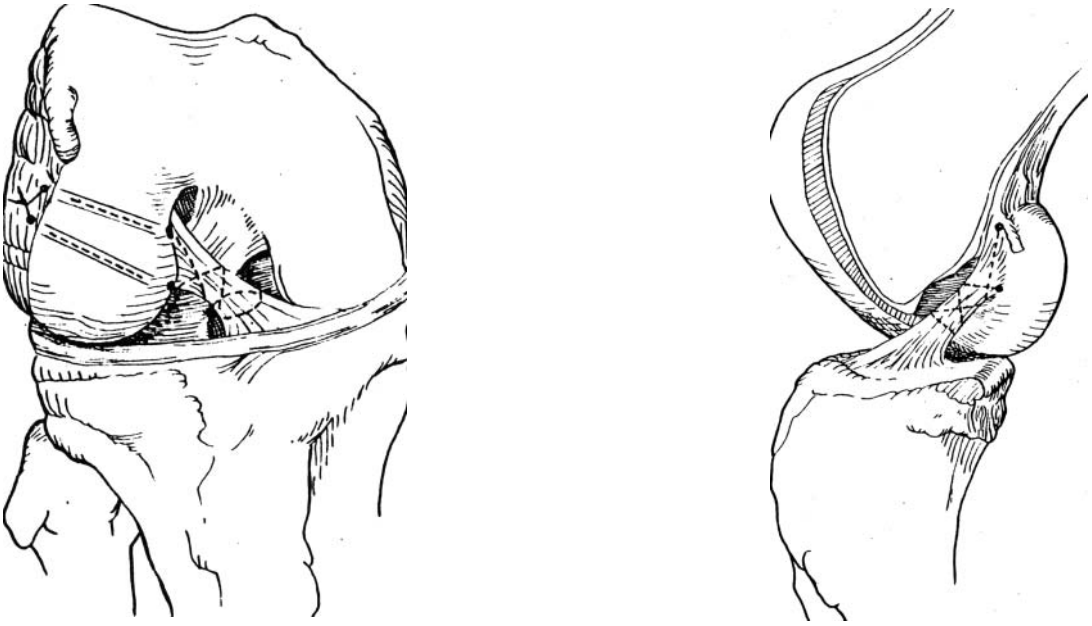


Figure 3. The proximal ACL tear is repaired to the femoral insertion through drill holes.

had a meniscal injury or other ligamentous injury identified at the time of the index arthrotomy.

Twenty-five of the 26 patients with complete ACL tears had a primary repair of the ACL. One patient with a complete midsubstance ACL tear had the ligament excised as it was considered nonrepairable. One patient had an end-to-end repair of a midsubstance ACL tear. Twenty-four had a proximal ACL repair as shown in Figure 3. A chromic or Mersilene suture was placed in a figure-eight fashion in the proximal portion of the torn ACL. The sutures were passed through 1 or 2 drill holes in the lateral femoral condyle. The sutures were tied over the iliotibial band in 18 cases and over a bony bridge in 6 cases. In the 8 patients with partial ACL tears, the torn portion was excised in 1, and in the other 7, the ligament was left alone. After the index operation, the patient's leg was placed in a long leg cast for approximately 6 weeks with the knee in 15° of flexion.

Evaluations included the subjective and symptom ratings of the International Documentation Committee (IKDC) evaluation form,[10] the Lysholm rating scale,[12] the Knee Injury and Osteoarthritis Outcome Score (KOOS),[16] the Tegner activity scale,[18] and SANE rating.[20]

We examined the data for any relationship with age at surgery, contact or noncontact injury, primary repair or no primary repair, partial tears or complete tears, subsequent operations, subsequent operations for meniscal injuries, subsequent operations for instability, 5-year SANE score, operative time, total number of days hospitalized, number of days hospitalized postoperatively, and time from injury to operation.

A multivariate logistical regression analysis was performed using SPSS (Chicago, Illinois) at Loyola University Medical Center in Maywood, Illinois. The Pearson product correlation statistic (r) was used for all correlations. Statistical significance was defined as $P < .05$.

RESULTS

Twenty of the 34 patients (71%) included in the study had at least 1 more knee procedure (range, 1-12 additional procedures) after the index operation. Of the 25 who underwent a primary repair of the ACL, 16 (64%) had at least 1 additional surgical procedure. Seven of that 25 (28%) had additional procedures, such as a pes anserinus transfer or patellar tendon ACL reconstruction, to address instability. Of the 8 patients with partial tears, 4 had additional knee surgery. One patient with a partial tear had 2 additional procedures designed to address instability.

The IKDC subjective and symptom results are shown in Table 1. There was nearly an equal split between those subjects who considered their result as normal or nearly normal (subjective score = 18, symptom score = 17) and those who considered their result as abnormal or severely abnormal (subjective score = 16, symptom score = 17). The Lysholm scores averaged 70.1 (range, 20-100) and are displayed in Figure 4.

The average SANE rating at 5 years for the 23 patients who were also in the Feagin and Curl[6] study was 74.8 (range, 49-110). The current SANE rating averaged 68.9 (range, 10-100). The distribution of SANE ratings is shown in Figure 5.

The overall KOOS rating for the 34 patients averaged 68.6 (range, 26.4-100). Figure 6 shows the distribution of scores for the 5 subcategories of the KOOS. One patient failed to complete the KOOS evaluation. The Tegner activity score averaged 3.7 (range, 1-7).

Poorer scores on some outcome measures were related to shorter overall hospital stays, subsequent operations, and subsequent operations for meniscal injury. There was also a significant correlation between the 5-year SANE and some of the outcome measures (Table 2). There were no

TABLE 1
International Knee Documentation Committee
Subjective, Symptom, and Overall Scores[a]

	Subjective	Symptom	Overall
A: normal	6	8 (1 TKR)	4
B: nearly normal	12 (1 TKR)	9	10 (1 TKR)
C: abnormal	7	10	10
D: severely abnormal	9	7	10

[a]TKR, total knee replacement.

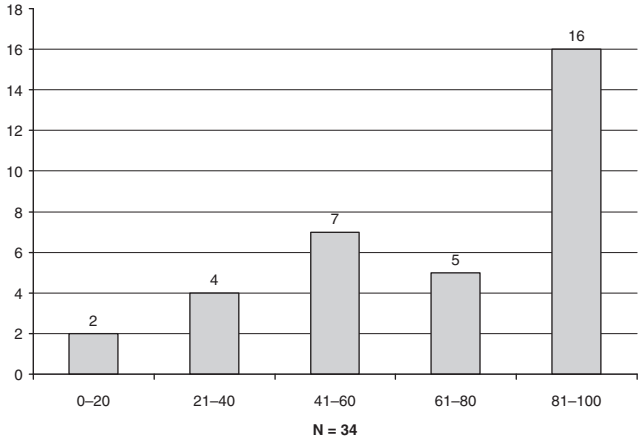


Figure 4. Distribution of Lysholm scores. Twenty-three of 34 patients (68%) had scores of 80 or better. The average score was 70.1.

significant correlations between the outcome measures with age at operation, contact injuries, primary repairs, partial tears, subsequent operations for instability, operative time, length of postoperative hospital stay, and time from injury to operation. The results for the partial tears were similar to the results for those that had complete tears and primary repair (Table 3).

Overall, there was a correlation between the SANE score and Lysholm score (r = .826, P < .0001) and between the SANE and total KOOS scores (r = .831, P < .0001). For KOOS subcategories, the SANE rating correlated best with the Quality of Life scale (r = .818, P < .0001) and least with the Function/Daily Living subcategory (r = .762, P < .001). The Lysholm correlated with the total KOOS score (r = .885, P < .0001)—best with the Pain subcategory (r = .875, P < .0001) and least with the Function/Sports and Recreational Activities subcategory (r = .786, P < .0001). Correlations between outcome measures are shown in Table 4.

DISCUSSION

For purely historical curiosity, today's surgeon should wonder about those patients who elected to undergo an unproven operation for a diagnosis that was difficult to make. Our experience with nonoperative treatment at

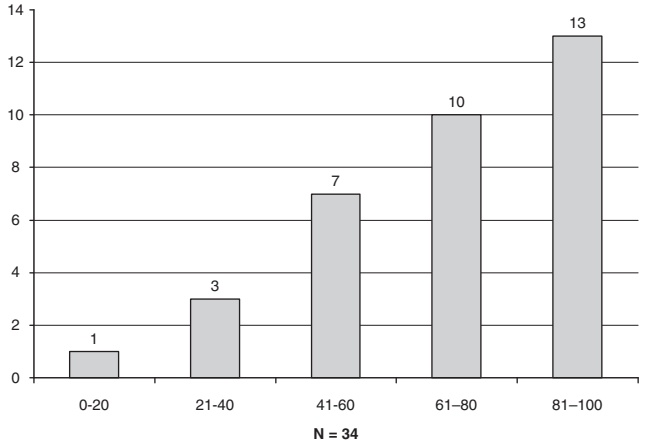


Figure 5. Distribution of Single Assessment Numeric Evaluation (SANE) scores. Twenty-one of 34 patients (62%) had scores of 80 or better. The average score was 68.9.

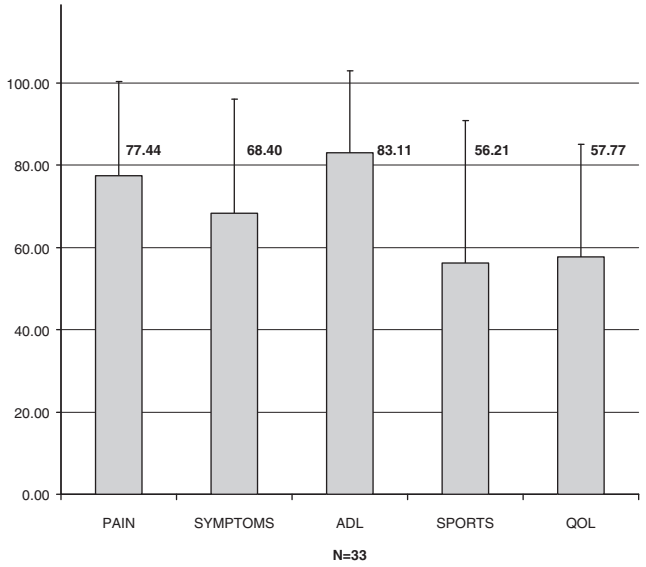


Figure 6. The scores for the Knee Injury and Osteoarthritis Outcome Score (KOOS) subscales. Values shown are the average score for each subscale. Error bars indicate standard deviation. ADL, activities of daily living; QOL, quality of life.

West Point had been similar to other natural history studies,[1,7,9,14,15] showing unacceptable results in a large percentage of patients. As a result of this experience, West Point surgeons during the 1960s and 1970s performed arthrotomies to identify ACL injuries and treat them with primary repair when possible. These operations led to the initial report that clarified the pathological and clinical findings associated with the noncontact "isolated" ACL injury and preliminary results of primary repairs.[5] This group of patients represents some of the earliest attempts at operative treatment of isolated ACL injuries in North America. Despite the promising 2-year results, it is clear that at 5 years, and now at greater than 30 years, the

TABLE 2
Significant Correlations[a]

	Additional Surgery	Meniscal Surgery	Total Days in Hospital	5-Year SANE Score
SANE	**$r = .474$, $P = .005$**	**$r = .584$, $P = .000$**	$r = .082$, $P = .656$	**$r = .420$, $P = .046$**
Lysholm	$r = .240$, $P = .171$	**$r = .367$, $P = .033$**	**$r = .374$, $P = .035$**	**$r = .540$, $P = .008$**
IKDC Symptom	$r = –.059$, $P = .740$	$r = –.111$, $P = .533$	**$r = –.374$, $P = .035$**	**$r = –.594$, $P = .003$**
IKDC Subjective	$r = –.260$, $P = .379$	**$r = –.415$, $P = .015$**	$r = –.154$, $P = .402$	**$r = –.452$, $P = .030$**
Tegner	$r = –.216$, $P = .227$	$r = .078$, $P = .664$	$r = .152$, $P = .415$	$r = .150$, $P = .494$
KOOS Pain	$r = –.059$, $P = .746$	**$r = .365$, $P = .037$**	$r = .271$, $P = .134$	**$r = .425$, $P = .049$**
KOOS Symptom	$r = .099$, $P = .584$	**$r = .416$, $P = .016$**	$r = .048$, $P = .795$	$r = .316$, $P = .152$
KOOS ADL	$r = .108$, $P = .551$	$r = .226$, $P = .206$	$r = .152$, $P = .406$	$r = .404$, $P = .062$
KOOS Sports	$r = .091$, $P = .615$	**$r = .445$, $P = .009$**	$r = .108$, $P = .556$	$r = .271$, $P = .223$
KOOS QOL	$r = .053$, $P = .768$	**$r = .439$, $P = .011$**	$r = .120$, $P = .513$	$r = .398$, $P = .067$

[a]SANE, Single Assessment Numeric Evaluation; IKDC, International Knee Documentation Committee; KOOS, Knee Injury and Osteoarthritis Outcome Score; ADL, activities of daily living; QOL, quality of life. Statistically significant correlations are in bold.

TABLE 3
Comparison of Subjective Outcomes for Anterior Cruciate Ligament Repairs Versus Partial Tears[a]

	Complete Tear With Repair (N = 25)	Partial Tear (N = 8)
SANE	68.5	70.0
Lysholm	71.2	70.3
IKDC Subjective	B	B
IKDC Symptom	C	B
Tegner	3.9	3.0
KOOS Pain	77.2	79.4
KOOS Symptom	68.1	69.9
KOOS ADL	83.1	85.1
KOOS Sports	57.6	57.1
KOOS QOL	58.5	58.0
5-year SANE	79.1	69.3

[a]SANE, Single Assessment Numeric Evaluation; IKDC, International Knee Documentation Committee; KOOS, Knee Injury and Osteoarthritis Outcome Score; ADL, activities of daily living; QOL, quality of life.

overall results of greater than 50% abnormal knees after repair of the ACL are unacceptable.

Many reconstructive techniques were developed after the publication of Feagin and Curl's 5-year follow-up study,[6] and enthusiasm for primary repair waned. Additionally, there have been many advances in knee surgery over the last 30 years, such as the development of arthroscopic techniques. Many of the techniques to treat ACL injuries have shown good short-term results, but longer follow-up has raised questions about deteriorating knee function over time. The results of this study provide baseline information for future long-term studies evaluating surgical treatment of ACL injuries.

One of the most interesting findings in our study was that the 5-year results were a good predictor of the subjective results at more than 30 years. The lower the SANE rating then, the poorer the subjective results 30 years later—one simple question was effective at determining how well patients do years later. It will be interesting to see if future long-term studies also have good correlation with midterm results, especially if the midterm results are good.

As with many long-term clinical studies, one of the biggest changes in our results over time was the decrease in activity level. As cadets, all patients were very active, rating at least a 7 on the Tegner activity score. Their activity levels remained high at 2- and 5-year follow-up. As one might expect, these patients' activity levels have declined after 30 years. The average Tegner activity score of 3.7 correlates with low impact recreational sports (swimming, cycling, cross-country skiing) and light to moderately heavy labor.

Not all patients had poor long-term results; in fact we found that nearly half of our patients did well. These good results may be related to the decreased activity levels and lower demands on the patients' knees, the ability of some patients to cope with ACL-deficient knees, or beneficial effects of the surgical arthrotomy with or without primary repair.

Others have reported good results with primary repair, such as Marshall et al[13] who reported good clinical results in 61 patients from the Hospital for Special Surgery (HSS) treated with primary ACL repair. At an average follow-up of 29 months, 93% of their patients were active in sports, and their average outcome score was 42.7 of 50. Seventy-five percent had a normal pivot-shift result, and 52% rated their knee as normal. These findings may appear similar to the 2-year results from our institution[5]; however, a subsequent HSS study,[11] with an average follow-up of 6 years and 10 months, found an average HSS score of 41.8 of 50. Fifty-eight percent had no abnormal laxity. These results, while not acceptable, were an improvement over the 5-year results reported by Feagin and Curl[6] and showed that primary repair did provide acceptable midterm results for some patients.

Sherman et al[17] reported on 50 primary ACL repairs with an average follow-up of 61 months. They found that 78% of their patients had a good to excellent result. The authors conducted an exhaustive statistical analysis to look for correlations between variables associated with ACL injury and the treatment methods used. Although no set of factors was definitively predictive of good results,

TABLE 4
Correlations Between Outcomes Measures[a]

	SANE	Lysholm	IKDC Symptom	IKDC Subjective	Tegner	KOOS Pain	KOOS Symptom	KOOS ADL	KOOS Sports	KOOS QOL
SANE		**r = .826, P = .000**	**r = –.670, P = .000**	**r = –.814, P = .000**	r = .262, P = .140	**r = .818, P = .000**	**r = .762, P = .000**	**r = .655, P = .000**	**r = .748, P = .000**	**r = .856, P = .000**
Lysholm	**r = .826, P = .000**		**r = –.865, P = .000**	**r = –.847, P = .000**	r = .304, P = .085	**r = .875, P = .000**	**r = .792, P = .000**	**r = .786, P = .000**	**r = .795, P = .000**	**r = .866, P = .000**
IKDC Symptom	**r = –.670, P = .000**	**r = –.865, P = .000**		**r = .730, P = .000**	r = –.279, P = .116	**r = –.711, P = .000**	**r = –.580, P = .000**	**r = –.600, P = .000**	**r = –.590, P = .000**	**r = –.680, P = .000**
IKDC Subjective	**r = –.814, P = .000**	**r = –.847, P = .000**	**r = .730, P = .000**		r = –.278, P = .117	**r = –.745, P = .000**	**r = –.786, P = .000**	**r = –.641, P = .000**	**r = –.766, P = .000**	**r = –.871, P = .000**
Tegner	r = .262, P = .140	r = .304, P = .085	r = –.279, P = .116	r = –.278, P = .117		r = .199, P = .274	r = .163, P = .373	r = .173, P = .342	r = .202, P = .267	r = .311, P = .083
KOOS Pain	r = .818, P = .000	**r = .875, P = .000**	**r = –.711, P = .000**	**r = –.745, P = .000**	r = .199, P = .274		**r = .866, P = .000**	**r = .900, P = .000**	**r = .815, P = .000**	**r = .828, P = .000**
KOOS Symptom	**r = .762, P = .000**	**r = .792, P = .000**	**r = –.580, P = .000**	**r = –.786, P = .000**	r = .163, P = .373	**r = .866, P = .000**		**r = .851, P = .000**	**r = .843, P = .000**	**r = .816, P = .000**
KOOS ADL	**r = .655, P = .000**	**r = .786, P = .000**	**r = –.600, P = .000**	**r = –.641, P = .000**	r = .173, P = .342	**r = .900, P = .000**	**r = .851, P = .000**		**r = .749, P = .000**	**r = .768, P = .000**
KOOS Sports	**r = .748, P = .000**	**r = .795, P = .000**	**r = –.590, P = .000**	**r = –.766, P = .000**	r = .202, P = .267	**r = .815, P = .000**	**r = .843, P = .000**	**r = .749, P = .000**		**r = .829, P = .000**
KOOS QOL	**r = .856, P = .000**	**r = .866, P = .000**	**r = –.680, P = .000**	**r = –.871, P = .000**	r = .311, P = .083	**r = .828, P = .000**	**r = .816, P = .000**	**r = .768, P = .000**	**r = .829, P = .000**	

[a]SANE, Single Assessment Numeric Evaluation; IKDC, International Knee Documentation Committee; KOOS, Knee Injury and Osteoarthritis Outcome Score; ADL, activities of daily living; QOL, quality of life. Statistically significant correlations are in bold.

there were characteristics that were associated with better outcomes. Overall, patients greater than 22 years of age had better results. Patients with proximal ACL avulsions had better results than those with midsubstance tears. A higher-grade preoperative pivot shift was associated with poor postoperative results. Finally, football injuries were associated with poor results, while skiing injuries were associated with proximal ACL avulsions and good quality of the remaining ligament. Their findings may help explain the poor results in the West Point studies. Consistent with the work of Sherman et al,[17] most cadets appear to be poor candidates for primary repairs, with an average age of 20 years and a large number of football-related injuries.

There has also been basic science evidence to support primary repair of the ACL. Cabaud et al[3] demonstrated in animal studies that the ACL has the capability to heal if adequately repaired. They found in rhesus monkeys that repaired ACLs appeared normal, restored knee stability, and resulted in an ACL that had approximately half the strength of the normal contralateral control. Augmented ACL repairs did even better, approaching normal strength.[3]

Despite these more positive studies investigating primary repair, our results fail to support the techniques of arthrotomy and primary repair for ACL injury, just as the 5-year results did. Furthermore, we were unable to identify any correlations at the time of injury that could help predict which patients would do well with primary repair. The correlations that we did identify were related to subsequent events and measures (subsequent operations, subsequent operations for meniscal injuries, and the 5-year follow-up SANE). We are unable to explain why longer overall hospital stays correlated with better results as measured by the significant correlations with the Lysholm and IKDC symptom scores.

Previous work has demonstrated that meniscal injuries influence the outcome of a patient treated for an ACL injury.[1,4,17] We add further evidence that the status of the menisci seems to be an important determinant in outcome after an ACL injury; lower outcome scores correlated with patients who had subsequent surgery for meniscal injuries. This finding may imply that the menisci were damaged due to subsequent injury to an unstable knee after the index procedure; however, another explanation may be that small to medium-sized tears were missed during the arthrotomy because visualization of the menisci was inadequate, especially by today's arthroscopic standards. Regardless of whether there was a meniscal tear with the initial injury or with subsequent injury, the status of the menisci was an important factor in the long-term outcome.

Notably, the partial tear group had a very similar result to the repair group. This similarity may be related to the difficulty in assessing ACL tears at the time of the original evaluation and operation. In the 1960s, the clinical assessment skills available were incomplete. The Lachman test[19] was not widely used, and magnetic resonance imaging (MRI) certainly was not available. The anterior drawer test and the history were the main criteria used to diagnose ACL injuries. Assessment of the ACL by arthrotomy was not as complete as an arthroscopic assessment. Additionally, the operations for some of the partial tears were performed after a delay—in 1 case, 30 days after injury. Thus, some of the

ACL injuries that were labeled as partial tears may have been complete tears that had partially healed. The combination of these factors suggests that the partial tears more likely represented a range of injuries from interstitial grade I sprains to complete ruptures, and this mixture of injuries in the partial tear group could explain the similarity of the results relative to the repair group.

Of note, a study by Barrack et al[2] compared the results of nonoperative treatment between partial and complete ACL tears identified arthroscopically. At an average follow-up of 41 months, they found that patients with partial ACL tears had better results and needed fewer reconstructions than those with complete tears. The partial tear group, however, had only 52% good or excellent results, which is similar to our long-term results for both the partial tear and primary repair patients.

There are difficulties interpreting the data collected over 30 years ago. All the records were hard copies and were well-maintained, but some of the information was incomplete and nonretrievable. Because of these difficulties, we included the information available for all the original 57 patients that we could contact and used the documents present in the research records as our source for the data on the index procedures and the 2- and 5-year follow-ups. Additionally, 60% follow-up is incomplete. However, those lost to follow-up were patients dispersed throughout the world that could not be located as opposed to patients unwilling to participate in this study. As a result, the patients that we were able to evaluate are more likely a good representative sample than some clinical studies with incomplete follow-up. Although the baseline information may not meet today's standards, we believe it does give us an adequate representation of this patient population as it was 32, 30, and 27 years ago.

CONCLUSION

For this patient population, the results at greater than 30 years reinforce the 5-year results that showed unsatisfactory results after the open evaluation and treatment of ACL injuries with or without repair. The major difference at final follow-up is that, as expected with increased age, our patients' activity levels have sharply decreased over time. We were unable to define any predictive factors to correlate with the results; however, subsequent meniscal surgery did correlate with poor results. The subgroup of patients with partial ACL tears that were not repaired had similar results to those with complete tears that had primary repairs. Despite the overall disappointing results after primary repair, approximately half of the patients considered their knees to be normal or nearly normal.

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